Annals of Botany **118**: 747–762, 2016
doi:10.1093/aob/mcw098, available online at www.aob.oxfordjournals.org

ANNALS OF BOTANY
Founded 1887

PART OF A SPECIAL ISSUE ON ROOT BIOLOGY

Anchorage failure of young trees in sandy soils is prevented by a rigid central part of the root system with various designs

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Received: 16 October 2015 Returned for revision: 28 January 2016 Accepted: 29 March 2016 Published electronically: 25 July 2016

• **Background and Aims** Storms can cause huge damage to European forests. Even pole-stage trees with 80-cm rooting depth can topple. Therefore, good anchorage is needed for trees to survive and grow up from an early age. We hypothesized that root architecture is a predominant factor determining anchorage failure caused by strong winds.
• **Methods** We sampled 48 seeded or planted *Pinus pinaster* trees of similar aerial size from four stands damaged by a major storm 3 years before. The trees were gathered into three classes: undamaged, leaning and heavily toppled. After uprooting and 3D digitizing of their full root architectures, we computed the mechanical characteristics of the main components of the root system from our morphological measurements.
• **Key Results** Variability in root architecture was quite large. A large main taproot, either short and thick or long and thin, and guyed by a large volume of deep roots, was the major component that prevented stem leaning. Greater shallow root flexural stiffness mainly at the end of the zone of rapid taper on the windward side also prevented leaning. Toppling in less than 90-cm-deep soil was avoided in trees with a stocky taproots or with a very big leeward shallow root. Toppled trees also had a lower relative root biomass – stump excluded – than straight trees.
• **Conclusions** It was mainly the flexural stiffness of the central part of the root system that secured anchorage, preventing a weak displacement of the stump. The distal part of the longest taproot and attached deep roots may be the only parts of the root system contributing to anchorage through their maximum tensile load. Several designs provided good anchorage, depending partly on available soil depth. Pole-stage trees are in-between the juvenile phase when they fail by toppling and the mature phase when they fail by uprooting.

Key words: Windthrow, 3D root architecture, *Pinus pinaster*, forest tree, biomechanics, toppling, tree anchorage, acclimation, soil depth, flexural stiffness, maximum tensile load.

INTRODUCTION

Anchorage and uptake of water and nutrients are the three primary functions of plant root systems. Anchorage has to be especially efficient in forest trees, because they grow tall stems to position their leaves to ensure optimal conditions for light availability against their competitors (Anten *et al*., 2005). Consequently, trees are vulnerable to strong winds, and because they live for a long time, the probability of them being hit by a storm is high. Thus, storms cause more than 50 % of primary damage to European forests (Schelhaas, 2008). One of the main types of damage is anchorage failure. When larger trees uproot, part of the surrounding soil remains entangled in the root system, so that a whole root–soil plate is lifted from the soil. Anchorage failure in smaller trees results in stem leaning because the central part of their root system is not stiff enough to make one common stiff element with the stem, and not big enough to hold a substantial volume of soil (Danjon *et al*., 2013*a*). Trees are considered as toppled if stem leaning exceeds 15° from the vertical (Mason, 1985). Even if a toppled tree does not die or require thinning, residual basal sweep will reduce wood quality (Burdett *et al*., 1986) and timber value (Cown *et al*., 1984). Young toppled trees will also be more likely to uproot thereafter (Coutts, 1983*a*; Danjon *et al*., 2005).

The vulnerability of a tree to strong wind depends on its aerial dimensions (stem and crown), and on root anchorage strength, which is in turn governed by soil properties, root architecture and root wood properties (Dupuy *et al*., 2005; Peltola, 2006). The contribution of broad root system components to anchorage was formerly studied by combining static winching and trenching in trees that failed by uprooting (e.g. Coutts, 1986; Nielsen, 1990) and in trees that failed by toppling (e.g. Crook and Ennos, 1996, 1998; Crook *et al*., 1997). Other types of root anchorage studies on young trees combined stem straightness or storm damage measurements and (1) a subjective scoring of shallow root distribution and taproot morphology (e.g. Mason, 1985; Gautam *et al*., 1999; Toral *et al*., 2011), (2) partial quantitative measurements generally including maximal rooting depth or root cross-sectional area (CSA) at a given radial distance and depth (Coutts, 1983*a*; Drexhage *et al*., 1999) or (3) detailed root architecture measurements (Danjon *et al*., 1999*b*, 2005). Structural root architecture measurement methods have been fully described (Danjon and Reubens, 2008). Lastly, biomechanical models of roots and soil have been used to describe the spatial and temporal distribution of

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TABLE 1. *Characteristics of the stands and trees sampled in March 2012*

	Seeding Full ploughing SFP	Planting Full ploughing PFP	Planting Rotary harrowing PRP	Planting Strip ploughing PSP
Tree age	13	14	14	14
Density (trees ha^{-1})	1348 (one pre-commercial thinning)	1376 (no thinning)	1399 (no thinning)	1554 (no thinning)
Mean height (m)	8·8	10·2	10·1	11·8
Mean dbh (cm)	14·1	15·8	15·9	16·8
% of toppled trees	18	30	38	33
		Sample		
Stem length end 2008 (m)	7	8·17	8·13	8·50
Number of straight trees	3 + 5*	3 + 6*	3	3
Number of leaning trees	3 + 1*	3	3	3
Number of toppled trees	3	3	3	3
Mean stem length (m)	9·7	10·7	10·6	10·9
Mean dbh (cm)	17·2	17·4	17	17·4
Tree ID	21–36	1–16	41–49	51–59

Ploughing depth was about 35 cm for all three ploughing conditions. dbh and stem leaning data were measured on 1250 trees, and height was measured on 276 trees. Stem length when hit by the storm was estimated from the position of the corresponding branch whorl on the stem. Trees 11–16 and 31–36 were winched trees.

*Trees from the concomitant winching study.

constraints, displacements, dynamics of stresses and failures during toppling or uprooting (Dupuy *et al*., 2005; Fourcaud *et al*., 2008; Yang *et al*., 2014).

Two main types of loadings affect roots as a function of their position in the structure, namely tension and flexion (Weber and Mattheck, 2005). Even if the mechanical properties of living coarse root wood largely influence anchorage, data on the spatial distribution of these properties are scarce (Yang *et al*., 2014).

The major components of a good anchorage of pole-stage trees have been examined in several studies. In 11-cm dbh (stem diameter at breast height) *Mallotus wrayi* trees grown in clay soil, the taproot contributed around 80 % of anchorage, and the six thin, 3-cm mean diameter shallow roots only 20 % (Crook *et al*., 1997). In 4·5-cm dbh *Pinus pinaster* Ait. saplings, trees with more pronounced stem leaning had a shallower rooting depth and a smaller deep root volume (Danjon *et al*., 1999*a*). In 3·5- to 12-cm dbh *Pinus radiata* grown in different soil types, the toppling probability decreased for trees with a straight vertical and dominant taproot, while the role of shallow roots was not really established (Toral *et al*., 2011). The stability of trees can be significantly reduced by an asymmetrical structure of shallow roots (Coutts *et al*., 1999), with missing roots or poorly developed roots windward (Coutts, 1983*b*). Conversely, root systems can exhibit large and selective acclimation to the prevailing wind (Danjon *et al*., 2005). The circular distribution of shallow roots in relation to prevailing wind azimuth was mainly studied by examining the position of the centre of mass of the entire root system or the CSA of shallow roots (e.g. Coutts *et al*., 1990; Quine *et al*., 1991; Nicoll *et al*., 1995), or using homogeneity indexes (e.g. Toral *et al*., 2011).

Pinus pinaster seedlings first grow a taproot and then lateral roots. Ten to 20 shallow horizontal roots develop from the stump (Burbidge, 1936). Secondary sinkers branch from these shallow roots and grow to more than 5-cm proximal diameter roots in large trees, forming a kind of cage around the stump that traps the soil and determines the volume of the root–soil plate in case of uprooting (Danjon *et al*., 2005). Trees in windy areas tend to develop a larger, unique and deep taproot (Danjon *et al*., 2009). Pines do not generally produce or regenerate new roots on the framework of undamaged original roots, at least to a structural extent (Burbidge, 1936; Atger and Edelin, 1994). Therefore, the anchorage of large *P. pinaster* trees depends strongly on the development of their root system as seedlings and saplings (Danjon *et al*., 2005, 2013*a*).

The first objective of this article is to present a new method for analysing root architecture by computing mechanical characteristics from morphological data. The second objective is to assess the influence of establishment techniques (seeding or planting; different types of ploughing) on root architecture and to determine which characteristics of the root system prevent the leaning or toppling of moderately deep rooted *P. pinaster* trees at the pole stage. We hypothesized that good anchorage is provided by a large, deep and vertical taproot, strong windward roots holding the tree in tension, and leeward roots with high flexural stiffness in the vicinity of the stump.

MATERIALS AND METHODS

The stands were located in the Landes de Gascogne Forest, an intensively managed *P. pinaster* forest that covers 1 million hectares in south-western France. The climate is temperate marine, with moderately warm summers and cool wet winters. Mean annual rainfall is 944 mm, mean annual temperature is 13.9 °C, and prevailing winds and storm winds blow from the west (1981–2010 data, Météo France, http://www.meteofrance.com/climat/; retrieved 11 February 2016). We selected one seeded and three planted neighbouring *P. pinaster* stands (Table 1). The stands were underlain by a medium humid sandy spodosol with a discontinuous deep hard pan at 40–90 cm depth (Augusto *et al*., 2010). The water table fluctuated close to the soil surface during rainy winters, but sank to 1.5 m depth in late summer. The four stands were damaged by Storm Klaus on 24 January 2009, when they were about 8 m high. Leaning and toppling azimuths were in the range 95±60° (Danquechin Dorval, 2015). An azimuth angle of 275° was therefore taken as the prevailing wind direction.

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In March 2012 (3 years after the storm), nine trees per stand were selected: straight trees (basal stem lean ≈3°), leaning trees (≈11°) and heavily toppled trees (≈30°). We sampled a total of 11 additional straight trees and one leaning tree in spring 2012 and 2013 (six trees each year) for a concomitant winching study (Yang *et al*., 2014) (Table 1). To choose trees at the same development stage, we sampled them within the narrow 15·9–17·8 cm dbh range and without any stem fork. The final sample size reached 48 trees. When leaning, *P. pinaster* trees grow larger tree rings in the leaning direction. Stem discs at breast height were therefore examined to determine when the selected trees started leaning.

The stem was felled, stem length was measured and root systems were uprooted using a combination of techniques (high-pressure air lance, hand tools, mechanical shovel) according to Danjon and Reubens (2008). We had to fix the root systems of the 12 winched trees with clamps and screws before digitizing them so as to recover the undamaged structure. We digitized the root systems according to Danjon *et al*. (1999b, 2005, 2013*b*), i.e. with each root axis subdivided into segments defined as the root part between two measurement points. We recorded the data in a multitree graph (MTG) format file (Godin *et al*., 1999). Roots less than 4 mm in diameter were only counted as a root number per segment. The dataset was composed of 9850 root axes and 52000 segments. We spent approximately 2.5 d per person and per tree for uprooting and digitizing.

Based on an architectural analysis (Barthélémy and Caraglio, 2007), we classified root axes into nine root-type classes according to Danjon *et al*. (2005, 2013*b*, *c*) and Danjon and Reubens (2008), using AMAPmod (Godin *et al*., 1999) (Fig. 1A). We considered the stump and the taproot as order 1. This first-order axis branched into order 2 roots and so on. When several vertical axes branched from the stump bottom, we classified the largest and most vertical one as the taproot and the others as second-order sinker roots. We called 'taproot complex' the taproot and all the sinker roots and oblique roots growing from the first-order axis. To cope with root system morphology, we set the lower limit for shallow roots and the upper limit for deep roots to respectively 25 cm and 66 % of the maximum rooting depth for each tree (Fig. 1A). The maximum rooting depth is generally determined by the hard pan or the water table level. The line running between the centre of the taproot origin and the point where the taproot crossed the 'deep root boundary' provided taproot vertical deviation. We set the boundaries between horizontal (shallow), oblique and vertical (sinker) roots to 30° and 60° towards the soil surface, respectively. A difference was made between sinkers branching from or beyond the zone of rapid taper (ZRT) because sinkers branching from the ZRT will shape the future cage at mature stage (Danjon *et al*., 2005). Specific root length was computed from the root volume as root length/root volume.

We made the following adaptations to better take into account the potential mechanical role of roots:

(1) **Relative variables** – As the critical turning moment in winching tests is tightly linked to stem volume (Cucchi *et al*., 2004; Nicoll *et al*., 2008), we divided all variables related to tree size by the tree stem volume, computed from stem length, ground base diameter and dbh. All the transformed variables were then referred to as 'relative' variables to link potential resistance of roots to the force they have to withstand (Nicoll *et al*., 2008).

(2) **Estimation of root volume lost during uprooting** – Almost all of the shallow roots were recovered, at least up to a 3-m radius from the stump, while the taproot and sinkers were generally gathered down to 1 cm diameter. However, some larger root parts were sometimes lost, for example when they grew under a neighbouring tree. Thus, to avoid a bias due to root loss, we estimated the loss for each root from its broken end diameter D_b, using regressions by root type between the cross-sectional diameters of 6000 selected segments and their descendant root volume V_d according to Danjon *et al*. (2013*c*). We then added a segment as an extension of each broken root in the MTG file with an 0.2-cm end diameter D_e and a length l_d computed as follows (Danquechin Dorval, 2015):

$$l_d = 12V_d/(p(D_b{}^2 + D_bD_e + D_e{}^2)).$$

(3) **Circular distribution of the shallow roots** – As in Danjon *et al*. (2013*b*), three discontinuous orientated sectors were defined, with a zero value for the prevailing wind direction: windward, (ww = 0±45°), leeward (lw = 180±45°) or perpendicular to the prevailing wind direction. However, these discontinuous sectors entail a threshold effect around the boundaries of each sector. To get rid of this threshold effect, we studied the weighted circular distribution of the root characteristics, assigning all segments except those of the first-order root to three discontinuous overlapping 180° sectors orientated according to the prevailing wind direction (Fig. 1C). The corresponding variables were weighted linearly between 1 and 0 according to the angle to the centre of the 180° sector. Instead of classifying the segments using the azimuth of the centre of the segment, we used the azimuth of the second-order roots at a 20-cm radial distance from the stump bark, from which the segments grew. In this way, the potential mechanical contribution of a segment was assigned to the azimuth where it was likely to apply.

We computed the size of the largest angle between shallow roots ('LABS'; Nielsen, 1990) in 180° sectors on both the leeward and the windward sides. All shallow roots with a diameter ≥3 cm at 20 cm from the stump bark were taken into account (Fig. 1C).

(4) **Computation of mechanical characteristics from morphological data** – We computed the CSA of roots as:

$$CSA = \pi D_1 D_2/4$$

where D_1 is one diameter and D_2 is the perpendicular diameter to D_1.

Maximum tensile load (Commandeur and Pyles, 1991) was defined as: *MOR* CSA, where *MOR* was the modulus of rupture and averaged 46 MPa in mature *P. pinaster* large roots (Défossez *et al*., 2014).

We defined the area moment of inertia I of an oval root in the direction of stress as:

$$I = \pi D_s^3 D_p/64$$

where D_s is the diameter in the stress direction, and D_p is the perpendicular diameter to the stress diameter. Flexural stiffness

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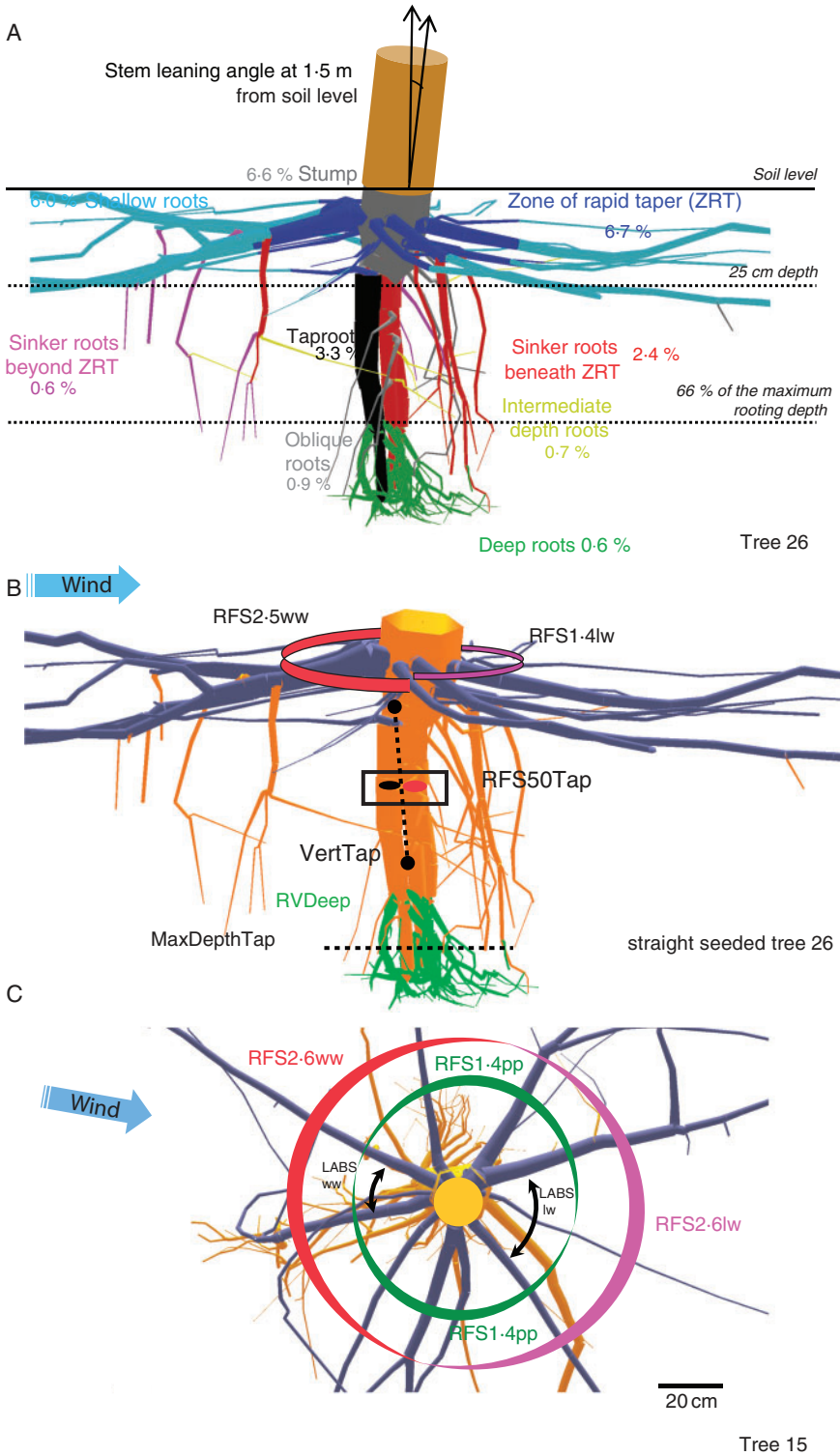


FIG. 1. (A) Scheme of the nine mechanical compartments and the spatial limits. Segments were coloured according to their compartment: (1) stump in grey, (2) taproot in black, (3) zone of rapid taper (ZRT) of horizontal shallow roots in dark blue, (4) horizontal shallow roots beyond the ZRT in light blue, (5) sinker roots branching from the ZRT in red, (6) sinker roots beyond the ZRT in magenta, (7) intermediate-depth horizontal roots in yellow, (8) deep roots in green, (9) oblique roots in dark grey. The mean percentage with respect to the stem volume in straight trees is indicated for each compartment. (B) Schematic representation of the variables used in the anchorage index: maximum taproot depth (maxDepthTap), taproot complex relative flexural stiffness at 50 cm depth (RFS50Tap), relative volume of deep roots below the ZRT (RVDeep), relative weighted flexural stiffness of leeward shallow roots at a radial distance of 1·4 dbh (RFS1.4lw) and relative weighted flexural stiffness of windward shallow roots at a radial distance of 2·6 dbh (RFS2.6ww). (C) Schematic representation of the largest angle between shallow roots (LABS) leeward (lw) or windward (ww) at 20 cm radial distance, and weighted relative flexural stiffness of shallow roots at a radial distance of 1·4 or 2·6 dbh. pp: perpendicular to wind. Shallow roots are in grey.

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or bending stiffness is the resistance of a structure that undergoes bending. The flexural stiffness of a beam at a given section was approximated by: $I\,E$, where E was the Young's modulus of the material (Tobin *et al*., 2007). We used a Young's modulus of 8 GPa to compute stiffness according to Stokes *et al.* (1997) and Khuder *et al*. (2007).

Mechanical characteristics such as maximum tensile load and flexural stiffness were computed using constant values for *MOR* and *E* whatever the root segment, because we did not have sufficient data to take into account their spatial variability in the *P. pinaste*r root system.

We computed estimations of relative maximum tensile load and of relative flexural stiffness to assess their spatial distribution in the root system in the following way:

a. for the taproot and the taproot complex, each centimetre between 30 and 75 cm depth;
b. for shallow roots, the weighted mean values in the windward and leeward sectors, from 0·2 to 6 dbh radial distance from the stump bark, by multiples of 0·1 dbh;
c. for sinkers branching from shallow roots in and beyond the ZRT, weighted values in the windward and leeward sectors, 2 cm from the base of the root because this zone is likely to experience high mechanical constraints.

We computed an 'ovality stiffening' variable as the ratio between original stiffness computed with $D_s{}^3D_p$ and stiffness computed for a round section with $D_s{}^2D_p{}^2$.

STATISTICAL ANALYSIS

As the root architectures of the trees from the three planted stands proved to be similar (Danquechin Dorval, 2015), they were pooled per leaning class and referred to as 'planted trees'. All the analyses were performed with R software (Ihaka and Gentleman, 1996). We computed a two-way analysis of variance with leaning class and regeneration method (seeding vs. planting) as fixed effects, and their interaction:

$$Y_{ijk} = \mu + \text{Leaning}_i + \text{Regeneration}_j + \text{Leaning}_i \cdot \text{Regeneration}_j + e_{ijk}$$

where Y_{ijk} is the value for tree k in level of leaning i, level of regeneration j, μ is the general mean and e_{ijk} is the residual error. In the case of non-normal residuals, we transformed variables to obtain a normal distribution.

To test if anchorage strength was linked to several features of root architecture, we defined an anchorage index from the following five variables (Fig. 1B) in the same way as Valdés-Rodríguez *et al.* (2013): (1) maximum taproot depth (maxDepthTap), (2) relative flexural stiffness of the taproot complex at 50 cm depth (RFS50Tap), (3) deep root relative volume (RVDeep), (4) relative flexural stiffness at 2·6 dbh radial distance of shallow roots on the windward side (RFS2·6ww) and (5) relative flexural stiffness at 1·4 dbh radial distance on the leeward side (RFS1·4lw) (Tables 2 and 3). The last variable was added because it was obviously a main anchorage component for three straight trees (nos. 24, 35 and 57 – Supplementary Data Table S1; Fig. 6). The anchorage index was calculated as the sum of all five variables after scaling each variable within the 0–1 range:

$$V_{\text{rescaled}} = (V - V_{\text{Min}})/(V_{\text{Max}} - V_{\text{Min}})$$

where V_{Min} is the minimum value and V_{Max} is the maximum value for each variable. The anchorage index ranged from 0 (probably a very unstable root system) to 6 (probably a very stable root system).

We also carried out a principal component analysis (PCA) of the 48 trees using the five scaled variables of the anchorage index.

RESULTS

Among the selected trees, differences between leaning classes were not significant regarding vertical crown area (6·2±1·3 m^2), or stem slenderness (stem length/dbh = 60·8±5·0). We therefore assumed homogeneity of the aerial parts. Examination of stem growth ring ovality at breast height indicated that toppled trees had started to lean when they were between 3 and 7 years old, but heavily toppled only as a consequence of the storm. Most of the leaning trees started leaning in their first years, except trees 41 and 43 which leaned due to the storm. The leaning and toppled trees leaned between 45° and 155° azimuth.

Planted trees possessed greater relative root volume (28·2 vs. 25·1 %, Table 2) due to a higher volume of stump (7·2 vs. 5·7 %) than seeded trees, and also higher relative weighted flexural stiffness of shallow roots at 2·6 dbh windward (12·8 vs. 6·65 kPa.m; Table 3, legend). Among the roots making up the taproot complex at 50 cm depth, both planted and seeded trees averaged four roots branching from the taproot, but seeded trees averaged only one root branching directly from the stump while planted trees had nearly 6 ($P = 2{\cdot}1\text{e-}05$, data not shown). Thus, planted trees possessed a higher number of roots in the taproot complex at 50 cm depth (10·8 vs. 5·7, Table 2). Conversely, planted trees had a lower taproot relative volume (2·64 vs. 3·47 %, Table 2), a lower maximal tensile load of the taproot at 50 cm depth (1410 vs. 2090 kPa.m^{-1}), and lower flexural stiffness of both the taproot and the taproot complex at 50 cm depth (91·7 vs. 166 kPa.m and 123 vs. 193 kPa.m, respectively). Planted trees did not differ from seeded trees as regards LABS or the other variables listed in Tables 2 and 3, anchorage index included. We found no interaction between leaning class and regeneration method, except for ovality stiffening (Tables 2 and 3).

Root volume, root depth and specific length

The leaning class effect was not significant for the total relative volume of the root system (around 27 % of the stem volume), but it was significant for the relative root volume, stump excluded (the *P*-value for the difference between straight and toppled trees reached 5·9 %), with large inter-tree variation ranging from 13·5 to 28·6 %. Among shallow-rooted trees (rooting depth less than 85 cm), all straight trees had a relative root volume (stump excluded) above 19 %, and all toppled trees were below that value. The lost roots

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TABLE 2. *Root systems of sampled trees: two-way analysis of variance of relative root volumes, and variables describing the taproot complex (*P *values, means±s.d.)*

Variable	Units	$P_{leaning}$	$P_{regeneration}$	$P_{interaction}$	Straight	Leaning	Toppling	Seeded	Planted
Relative root volume, stump included	%	0·067	0·024	0·85	27·9±4·1	28·4±4·74	24·9±4·81	25·1±4·86	28·2±4·22
Relative root volume, stump excluded	%	0·037	0·14	0·93	21·3*±3·08	21·4*±3·69	18·3*±4	19·4±3·94	21·1±3·49
Relative stump volume	%	0·69	0·0037	0·76	6·61±1·89	7·02±1·71	6·57±1·61	5·69±1·36	7·18±1·73
Relative tap root volume	%	0·024	0·014	0·34	3·32a±1·39	2·89ab±1·13	2·11b±0·932	3·47±1·03	2·64±1·32
Relative volume of taproot complex	%	0·014	0·18	0·71	5·55a±1·43	4·9ab±1·66	4·01b±1·28	4·56±1·22	5·18±1·69
Relative deep root volume	%	0·0071	0·053	0·36	0·603a±0·425	0·606a±0·313	0·215b±0·127	0·648±0·49	0·443±0·294
Maximum depth	cm	0·052	0·093	0·35	−96·9±11·7	−91·9±12·1	−88·2±6·67	−89·5±9·87	−95·2±11·6
Maximum depth of taproot	cm	0·022	0·28	0·44	−87·7b±11·5	−83·7ab±14·4	−76·3a±6·71	−81·1±9·88	−85±13
Taproot vertical deviation	degrees	0·24	0·07	0·8	9·94±8·55	12·9±13·4	15·7±10·2	7·92±4·35	14·1±11·8
Specific root length, first-order root excluded	cm/cm^{-3}	0·02	0·64	0·85	0·932b±0·31	0·906b±0·16	1·24a±0·411	0·969±0·249	1·02±0·364
Maximum tensile load of taproot at 50 cm depth	$kPa.m^{-1}$	0·00057	0·0031	0·18	2030a±799	1560ab±747	896b±792	2090±830	1410±830
Flexural stiffness of taproot at 50 cm depth	kPa.m	0·00044	0·0059	0·2	160a±126	97·5ab±75·5	46·4b±85·4	166±134	91·7±94
Maximum tensile load of taproot complex at 50 cm depth	$kPa.m^{-1}$	0·0055	0·91	0·78	3580a±1020	2920ab±1220	2270b±913	3100±937	3070±1260
Flexural stiffness of taproot complex at 50 cm depth	kPa.m	0·00026	0·014	0·42	198a±122	129ab±86·8	61·1b±80·6	193±128	123±104
Number of roots of taproot complex at 50 cm depth	*n*	0·65	0·0085	0·6	8·74±6·02	8·69±5·98	10·8±7·21	5·7±4·27	10·8±6·35
Ovality stiffening of taproot at 50 cm depth	*n*	0·7	0·48	0·42	0·992±0·105	0·963±0·0883	0·984±0·109	0·967±0·121	0·989±0·0906
Ovality stiffening of taproot complex at 50 cm depth	*n*	0·87	0·89	0·8	0·982±0·912	0·964±0·0833	0·983±0·145	0·981±0·146	0·976±0·0796

Different letters indicate significant differences using Tukey's multiple comparison test with significance threshold of 5 %· 'Relative' means divided by the stem volume.
*Tukey's multiple comparison test was not significant for individual comparisons.

TABLE 3. *Root systems of the sampled trees: two-way analysis of variance of the circular distribution of shallow roots and secondary sinkers (*P *values, means±s.d.)*

Variable	Units	$P_{leaning}$	$P_{regeneration}$	$P_{interaction}$	Straight	Leaning	Toppling
LABS at 20 cm radial distance windward	degrees	0·57	0·93	0·57	123±53	119±30·4	106±38·3
LABS at 20 cm radial distance leeward	degrees	0·99	0·31	0·34	74·1±18·1	73·7±25·1	72·5±21·2
Relative weighted flexural stiffness at 2·6 dbh ww	kPa.m	0·037	0·021	0·45	13·7±10·6	9·85±8·98	6·46±3·85
Relative weighted flexural stiffness at 1·4 dbh lw	$kPa.m^{-1}$	0·54	0·35	0·95	70·2±65·3	59·7±42·5	51±56·5
Relative weighted flexural stiffness at 1·4 dbh hpp	kPa.m	0·094	0·92	0·73	40·9±24	35·8±13·9	26±20·9
Relative weighted flexural stiffness at 2·6 dbh hpp	kPa.m	0·077	0·078	0·64	15·6±8·75	15·7±6·1	10·8±10·3
Weighted number of roots at 2·6 dbh ww	%	0·0077	0·75	0·85	13·8 b ± 6·52	14·3 b ± 7·45	22·3 a ± 8·98
Number of 2nd order roots in ZRT	*n*	0·35	0·15	0·44	18·2±6·46	18·2±4·66	21·3±5·55
Number of 3rd, 4th and 5th order roots in ZRT	*n*	0·0013	0·95	0·39	5·26 b ± 5·62	9·31 a ± 7·85	14·2 a ± 11·1
Ovality stiffening at 2·6 dbh ww	%	0·00035	0·43	0·031	0·946 b ± 0·0392	0·953 b ± 0·0415	1·02 a ± 0·0839
ww/(lw + ww) ratio for maximum tensile load at 1·4 dbh	%	0·84	0·24	0·9	0·575±0·162	0·604±0·193	0·57±0·103
ww/(lw + ww) ratio for flexural stiffness at 1·4 dbh	%	0·79	0·082	0·9	0·639±0·227	0·679±0·279	0·675±0·209
Relative weighted flexural stiffness of secondary sinkers below ZRT ww	kPa.m	0·23	0·32	0·75	2.38±4·94	20·4±65·4	11·8±26·7
Relative weighted flexural stiffness of secondary sinkers below ZRT lw	kPa.m	0·28	0·61	0·53	21·6±40·9	4·12±6·85	5·73±10·4
Anchorage index (based on five variables)		9e-07	0·35	0·98	1·78 a ± 0·4	1·45 a ± 0·528	0·829 b ± 0·244

Different letters indicate significant differences using Tukey's multiple comparison test. 'Relative' means divided by the stem volume. 'Weighted' means weighting linearly between 1 and 0 according to the angle to the centre of the ±90° sector of a given direction. The relative weighted flexural stiffness at 2·6 dbh windward averages 6·65±5·27 kPa.m in the seeded trees and 12·8±10·2 kPa.m in the planted trees. LABS, largest angle between shallow roots; ww, windward; lw, leeward; pp, perpendicular to wind; dbh, diameter at breast height; ZRT, zone of rapid taper.

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averaged 19 % of the shallow root beyond the ZRT volume, 14·5 % of the deep root volume and 7·6 % of the root system volume, stump excluded.

In straight trees, the stump, taproot, shallow horizontal roots and sinkers represented approx. 7, 3, 13 and 3 % of the stem volume, respectively (Fig. 1A). The remaining root types made up less than 2 % of the stem volume. Among the relative volume variables of the nine root compartments, the leaning class effect was only significant for the taproot/taproot complex relative volume (3·3/5·5 % for straight trees vs. 2·1/4 % for heavily toppled trees, Table 2) and for the deep root relative volume (0·6 % for straight and leaning trees vs. 0·2 % for heavily toppled trees). The stump relative volume was highly variable (4–11 %).

The difference between straight and toppled trees was not significant for maximal rooting depth, but it was significant for taproot depth (88 vs. 76 cm, respectively). The difference between maximal rooting depth and taproot depth averaged 10 cm, but reached 25 cm for certain trees. No significant difference was evidenced for taproot vertical deviation. Specific root length, order 1 roots excluded, was significantly greater in toppled trees, averaging 1·24 vs. 0·93 for straight trees (Table 2).

Taproot

Differences between straight and toppled trees as to the mechanical characteristics of the taproot and taproot complex were significant at practically all depths (Fig. 2A). However, highly significant differences were only found for flexural stiffness in an area located between 47 and 65 cm depth, both for the taproot and for the taproot complex. Flexural stiffness of the taproot and taproot complex at 50 cm depth was respectively 3·5 and 3·2 times greater in straight trees than in toppled trees. None of the 30 % of trees of the sample with a taproot growing below 90 cm depth toppled, and only one tree (no. 57) among the 15 % of trees with less than 75 cm rooting depth of the taproot remained straight (Fig. 3). In the same vein, only one toppled tree (tree 21, see Fig. 6) had a taproot relative volume greater than 3 % (Fig. 3). Trees with less than 2 % taproot relative volume may have a high probability of leaning or toppling (9/11). The taproot alone made up 54 % of the maximum tensile load of the taproot complex, but up to 76 % of the flexural stiffness of the taproot complex in straight trees vs. 55 % in toppled trees ($P = 0{\cdot}02$).

The deep root volume was correlated (0·64, $P = 7{\cdot}5\text{e-}07$) to taproot flexural stiffness at 50 cm depth, but not to taproot depth (Table 4). The deeper roots of the five trees with a taproot depth greater than 97 cm did not exhibit the typical morphology due to the hard pan; that is, when the taproot or the sinkers reach the hard pan, they produce several dozen roots running horizontally just above it, and are progressively grafted to form 'hard pan shields' (Danjon *et al.*, 2005).

Shallow roots

There was no difference among leaning classes in any sector or at any radial distance (0·2–6 times dbh) as regards the relative maximum tensile load of shallow roots (Fig. 2A). Their relative flexural stiffness was only significantly different on the windward side between 1·7 and 2·6 dbh radial distance from the stump (approx. 30–45 cm; Fig. 2A; Table 3). Half of the straight trees and only one toppled tree had a relative flexural stiffness at 2·6 dbh radial distance greater than 12 kPa.m on the windward side. Contribution to stiffness of the vertical root ovality (ovality stiffening) of these roots at 2·6 dbh radial distance was more pronounced in straight and leaning trees than in toppled trees (+ 5 % mean stiffness vs. + 0 % in toppled trees, Table 3). The largest angles between shallow roots were not correlated to the corresponding computed mechanical characteristics (Table S1). Acclimation of shallow roots at 1·4 dbh radial distance resulted in a 42:58 windward/leeward ratio for maximum tensile load, and in a 35:65 ratio for flexural stiffness. These ratios did not differ among leaning classes (Table 3).

In all three leaning classes, trees had approx. 20 order 2 shallow roots branching from the stump. However, toppled trees averaged at least 14 shallow roots branching from the ZRT, vs. five for straight trees.

The mechanical characteristics at the base of the secondary sinker roots in the ZRT or beyond the ZRT were extremely variable, and no significant difference was found among leaning classes. The five trees with the greatest flexural stiffness of leeward sinkers in the ZRT were straight trees. Leaning tree 55 had a 9-cm basal diameter secondary sinker windward, scoring 27 times the mean flexural stiffness for this root component; it had probably counterbalanced a 100° leeward circular sector with almost no root.

The few trees with an exceptionally great flexural stiffness of their taproot complex did not display an exceptionally great flexural stiffness of their windward or leeward shallow roots. The few trees with an exceptionally great flexural stiffness of their leeward shallow roots did not have exceptionally stiff windward shallow roots, and vice versa (Table S1).

Anchorage index and PCA

The anchorage index pointed to a highly significant leaning class effect (Table 3). All straight trees had an index greater than 0·98, and the average value was 1·8±0·4 (Fig. 4). Tree 34 (Fig. 6) reached 2·7. Conversely, the index of toppled trees scored an average 0·8±0·2. The anchorage index was correlated (0·52, $P = 0{\cdot}0002$) with the relative root volume, stump excluded: this correlation was not significant (0·34, $P = 0{\cdot}1$) among straight trees (Table 4). The anchorage index was mostly correlated to the flexural stiffness of the taproot complex and to the deep root relative volume.

On the graph of the first two components of the PCA (PC1 and PC2), which explained 62 % of the variance (Fig. 5), all toppled trees except one were grouped toward low positive loadings for PC1 and PC2, whereas straight trees were spread across the remaining part of the graph, similarly to the five anchorage index variables positively associated with anchorage. PC1 was mainly associated with a low flexural stiffness of the taproot complex and a low deep root relative volume, and differentiated toppled trees from straight trees. PC2 was mainly associated with a low flexural stiffness of

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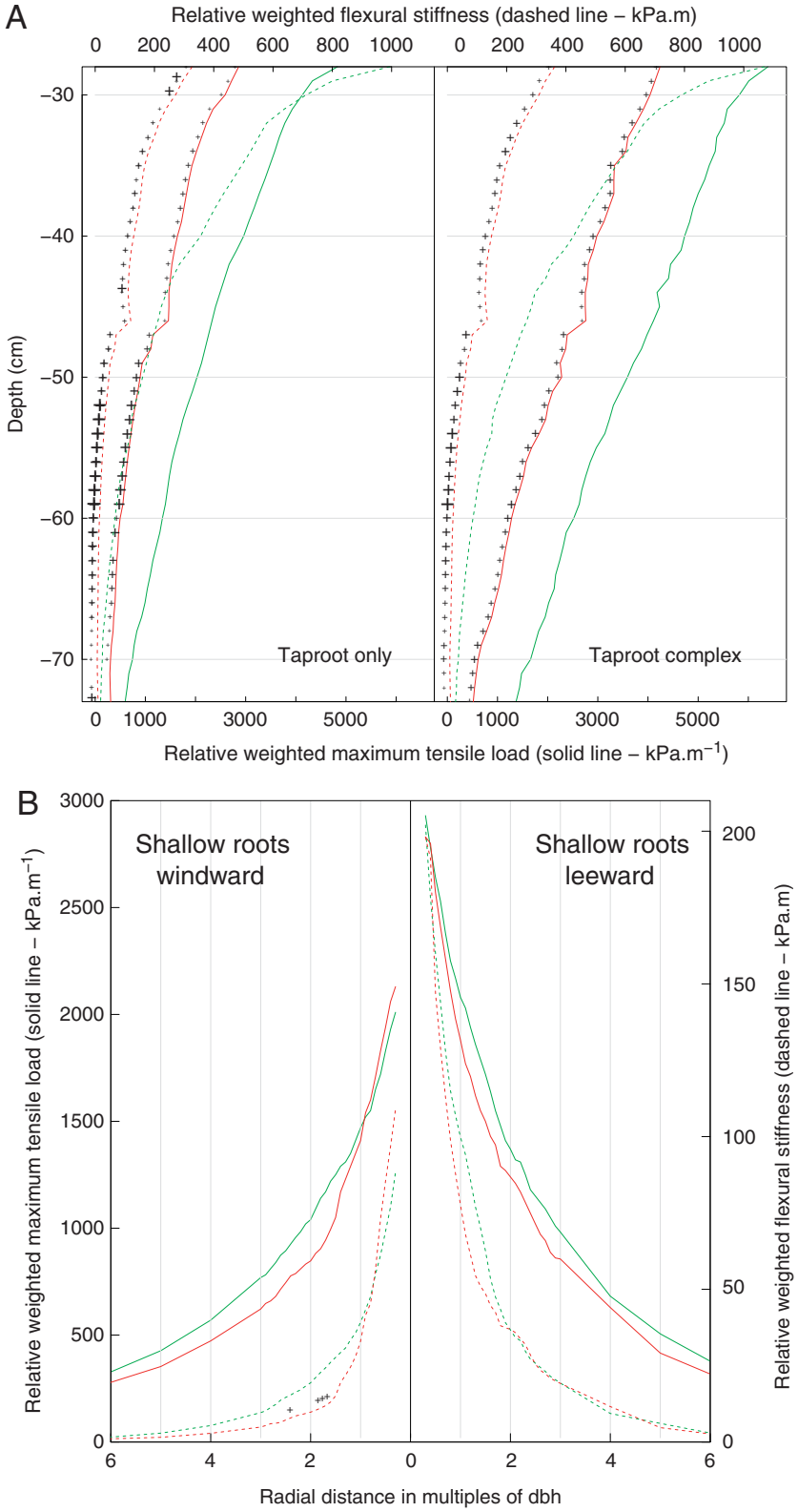


FIG. 2. Mean values of relative maximum tensile load (solid line) and relative flexural stiffness (dashed line) computed from morphological data for straight (green) and heavily toppled (red) trees. (A) Values for the taproot (left) and the taproot complex (right) as a function of soil depth (in cm). Size of black crosses is proportional to Tukey's HSD test for differences between straight and heavily toppled trees (computed as $-\log_{10}(P_{\text{tuckey}})$ when $P < 0{\cdot}05$; the largest '+' is for the highly significant difference 3e-06, the smallest '+' for $P = 0{\cdot}045$). (B) Values for shallow roots, with variation computed as a function of the radial distance from the stump bark, in multiples of tree dbh (6 is approximately $17 \times 6 = 100$ cm). Differences between standing and toppled trees are only significant for flexural stiffness between 1.8 and 2.6 dbh radial distance windward in the [0·1, 12] dbh range (the largest '+' is for $P = 0{\cdot}037$, the smallest for $P = 0{\cdot}047$).

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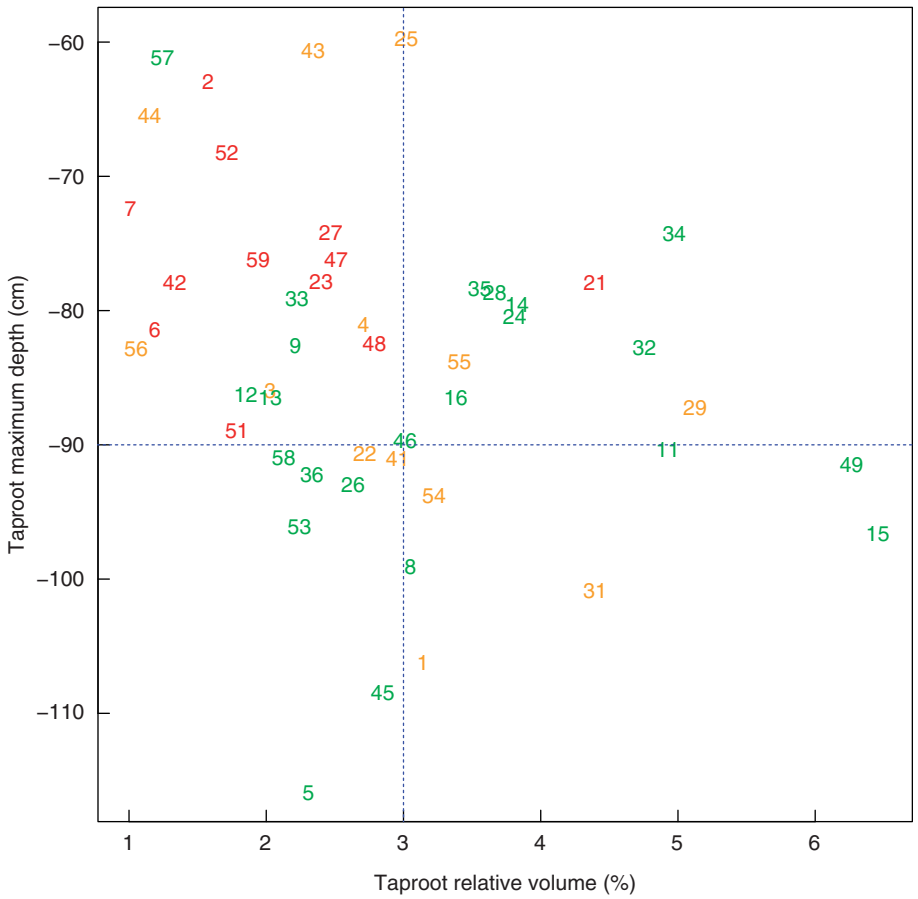


FIG. 3. Maximum taproot depth against relative taproot volume. Green numbers: straight trees, orange: leaning trees, red: toppled trees. Dotted blue lines define maximum values for toppled trees, except for tree 21.

TABLE 4. *Correlations between the five variables of the anchorage index plus the relative root volume without stump*

	STrootVol2	maxDepthTap	RFS50TRC	RVDeep	RFS2.6ww	RFS1.4lw	Ai
Relative root volume, stump excluded		−0.17 (0.435)	−0.03 (0.876)	0.32 (0.132)	0.47 (0.0237)	−0.04 (0.842)	0.34 (0.107)
Maximum depth of taproot	0.18 (0.213)		−0.11 (0.615)	−0.28 (0.195)	−0.22 (0.315)	−0.48 (0.0212)	−0.22 (0.311)
Flexural stiffness of taproot complex at 50 cm depth	0.17 (0.242)	0.23 (0.124)		0.61 (0.00197)	−0.25 (0.247)	−0.12 (0.598)	0.68 (0.000328)
Relative deep root volume below ZRT	0.37 (0.0107)	0.04 (0.763)	0.64 (7.5e−07)		−0.08 (0.724)	−0.08 (0.703)	0.77 (1.84e−05)
Relative weighted flexural stiffness at 2.6 dbh ww	0.41 (0.0036)	−0.01 (0.928)	0.04 (0.804)	0.1 (0.482)		−0.26 (0.233)	0.09 (0.7)
Relative weighted flexural stiffness at 1.4 dbh lw	0.15 (0.32)	−0.23 (0.114)	−0.03 (0.862)	−0.02 (0.874)	−0.2 (0.174)		0.22 (0.314)
Anchorage index	0.51 (0.000193)	0.38 (0.00711)	0.75 (9.59e−10)	0.73 (5.34e−09)	0.32 (0.0252)	0.28 (0.0506)	

Below the diagonal: all trees; above the diagonal: straight trees only. *P*-values are given in parentheses. Abbreviations: ww, windward; lw, leeward; dbh, diameter at breast height; ZRT, zone of rapid taper.

the leeward shallow roots and, to a lesser extent, positively correlated with a greater taproot depth and greater flexural stiffness of the windward shallow roots. Five straight trees out of six with less than 90 cm maximum rooting depth were located at the bottom of the graph, with a PC2 loading lower than −0·3.

The greatest values by far of shallow root flexural stiffness leeward were recorded for the four straight trees at the bottom left. They had all grown one very thick (10–15 cm diameter) oval shallow root leeward, except tree 57 which had three shallow roots 6–12 cm in diameter leeward. Nine toppled trees out of 12 had less than 90 cm maximum rooting depth, and also had

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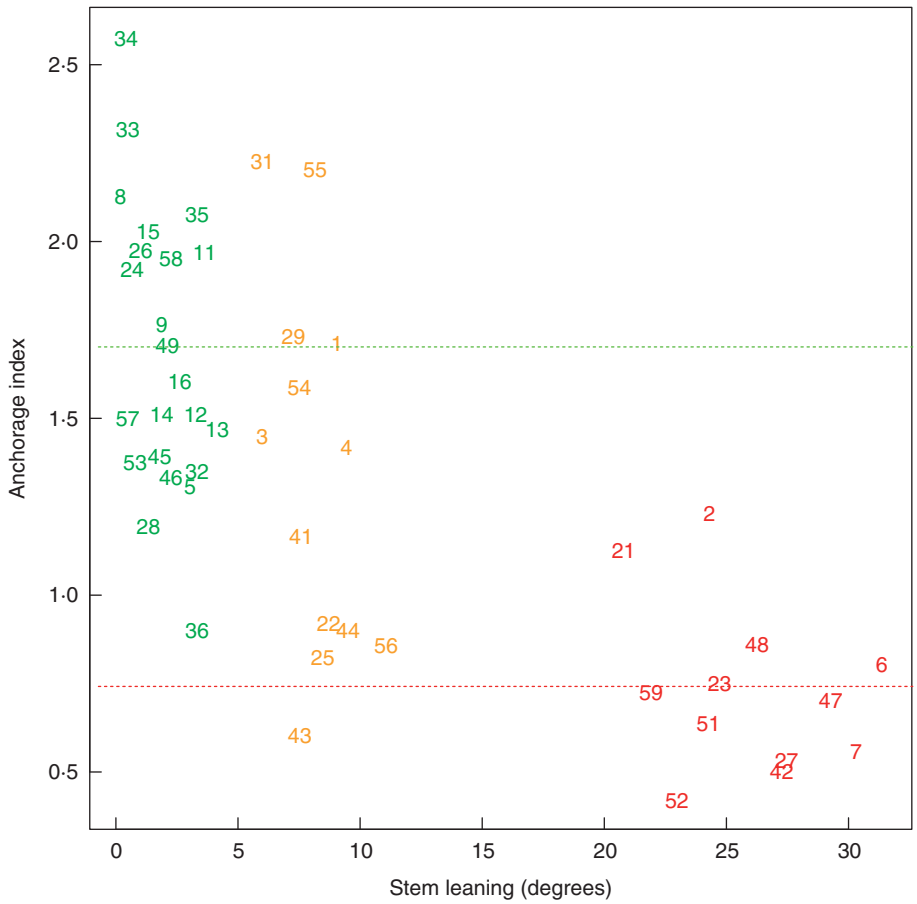


FIG. 4. Stem leaning against anchorage index. Green numbers: straight trees, orange: leaning trees, red: toppled trees. Dotted green line: mean anchorage index for straight trees, dotted red line for toppled trees.

low flexural stiffness leeward, except tree 2 situated at the bottom of the PCA graph (Fig. 5).

For most of the variables and the PCA, leaning trees had intermediate characteristics between straight and toppled trees. However, leaning trees were grouped with straight trees for anchorage index and for relative deep root volume. And the two trees that started leaning during the storm (nos. 41 and 43) were grouped with toppled trees in the PCA.

DISCUSSION

There was no fundamental difference in anchorage functioning between planted and seeded trees

Unexpectedly, we did not find a clear influence of soil preparation on the root architecture characteristics we studied. In sandy soils, ploughing probably mainly modifies the vigour of the understorey and the microtopography of the soil, and these do not appear to influence the distribution per compartment or the circular distribution of root characteristics.

Planted trees had a thicker stump and a thinner taproot than seeded trees. In the nursery, the taproot is air-pruned or pruned before planting. After planting, while the stump grew thicker, the newly formed taproot did not catch up on growth. This altered allocation was still visible in our 17-cm-dbh trees. When *P. pinaster* root growth is not disturbed, roots branching from the stump grow as horizontal shallow roots (Burbidge, 1936). Oblique and sinker roots branching from the stump of seeded trees were probably originally secondary sinkers whose base was no longer visible because of the stump secondary growth. Conversely, root deformation during nursery growth or at planting time probably gave rise to the additional roots contributing to the taproot complex in planted trees (higher number of roots in the taproot complex; Table 2). These additional roots offset the lower maximal tensile load of the taproot in planted trees but did not offset its lower flexural stiffness. Deformations did not affect the flexural stiffness of shallow roots, whether windward or leeward. The lower flexural stiffness of the taproot in planted trees was compensated for by greater flexural stiffness of the windward shallow roots. Unlike what is generally reported in the literature (Halter *et al.*, 1993; Lindström and Rune, 1999), the circular distribution of shallow roots measured from the LABS was the same in planted trees and in seeded trees. We found no interaction between stem leaning and regeneration method, which means that there was no fundamental difference in anchorage functioning between planted and seeded trees of the same size.

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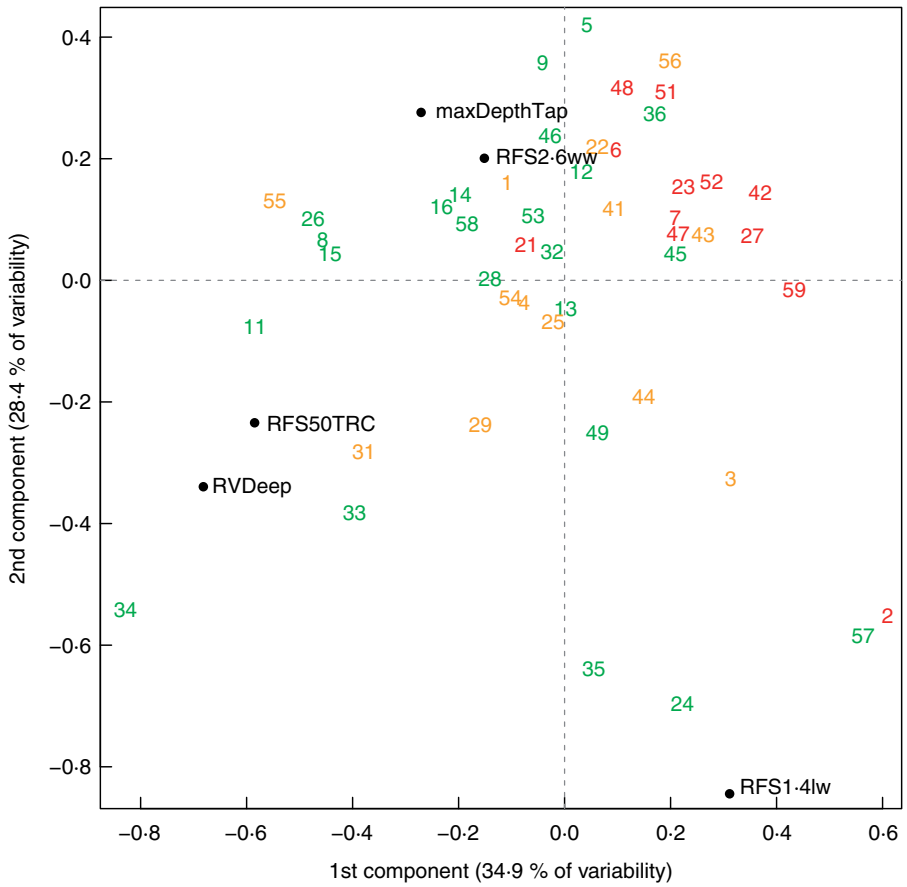


FIG. 5. Principal component analysis (PCA) of 48 trees using the five scaled variables composing the anchorage index: taproot depth (maxDepthTap), taproot complex relative flexural stiffness at 50 cm depth (RFS50TRC), deep root below ZRT relative volume (RVDeep), weighted relative flexural stiffness of windward shallow roots at a radial distance of 2·6 dbh (RFS2·6ww) and weighted relative flexural stiffness of leeward shallow roots at a radial distance of 1·4 dbh (RFS1·4lw). Green numbers: straight trees, orange: leaning trees, red: toppled trees.

Several components of the root system prevent toppling

A broad array of root system designs appears to be effective in maintaining trees straight. As evidenced by the anchorage index ranking and the PCA, anchorage strength in these young trees was multifactorial, as noted by Coutts (1983*a*) and Danjon *et al.* (2005) in larger trees. The mean anchorage indexes of straight and toppled trees differed only by 1 point. This corresponds to the full range of variation of one of the five variables. Toppled trees scored generally poorly for all the components of the anchorage index, whereas straight trees often scored highly for a couple of components. Leaning trees generally had intermediate characteristics between straight and toppled trees. The highest anchorage index value was only 2·7/5 (seeded tree 34, Fig. 6), and therefore in our sample no tree displayed a high score in all five desirable traits. During a storm, wind strength and direction can vary considerably within a stand because of several parameters such as the distance to the edge, landscape fragmentation or tree size distribution (Dupont *et al.*, 2015). Trees are often damaged by gusts of wind, which are more frequent inside the stand and blow in a broad range of directions. Therefore, the wind load on trees within a stand is far from being uniform, and could explain why some trees with a small index score did not topple.

Lower volume allocation to the roots, stump excluded, in toppled trees partly explained anchorage failure, similarly to *P. sitchensis* clones (Nicoll *et al.*, 1995) and to mature *P. pinaster* trees (Danjon *et al.*, 2005). The correlation pattern between the anchorage index and the relative root volume indicated that the rather low index values – even for straight trees – were only partly due to differential allocation between anchorage components.

The taproot is the main component of root anchorage in the juvenile phase

All the trees possessed what we could classify as a taproot. More than 90 cm taproot depth or more than 3 % taproot relative volume was sufficient to prevent toppling in almost all cases, whereas trees with less than 75 cm taproot depth or below 2 % taproot relative volume had a high probability of leaning or toppling. All the toppled trees except tree 21 had a weak taproot complex. The taproot complex, particularly the section located between 47 and 63 cm depth, was the main component of anchorage in these trees up to the pole stage. During winching, the centre of rotation of the taproot of *M. wrayi* trees of similar size and structure was located at 50 cm depth (Crook *et al.*, 1997). In our trees, the rigid part of the taproot acted as a stake in the soil (Ennos, 1993), anchoring the tree into the ground. Within the taproot complex, the taproot was dominant. Differentiation between straight and toppled trees was greater

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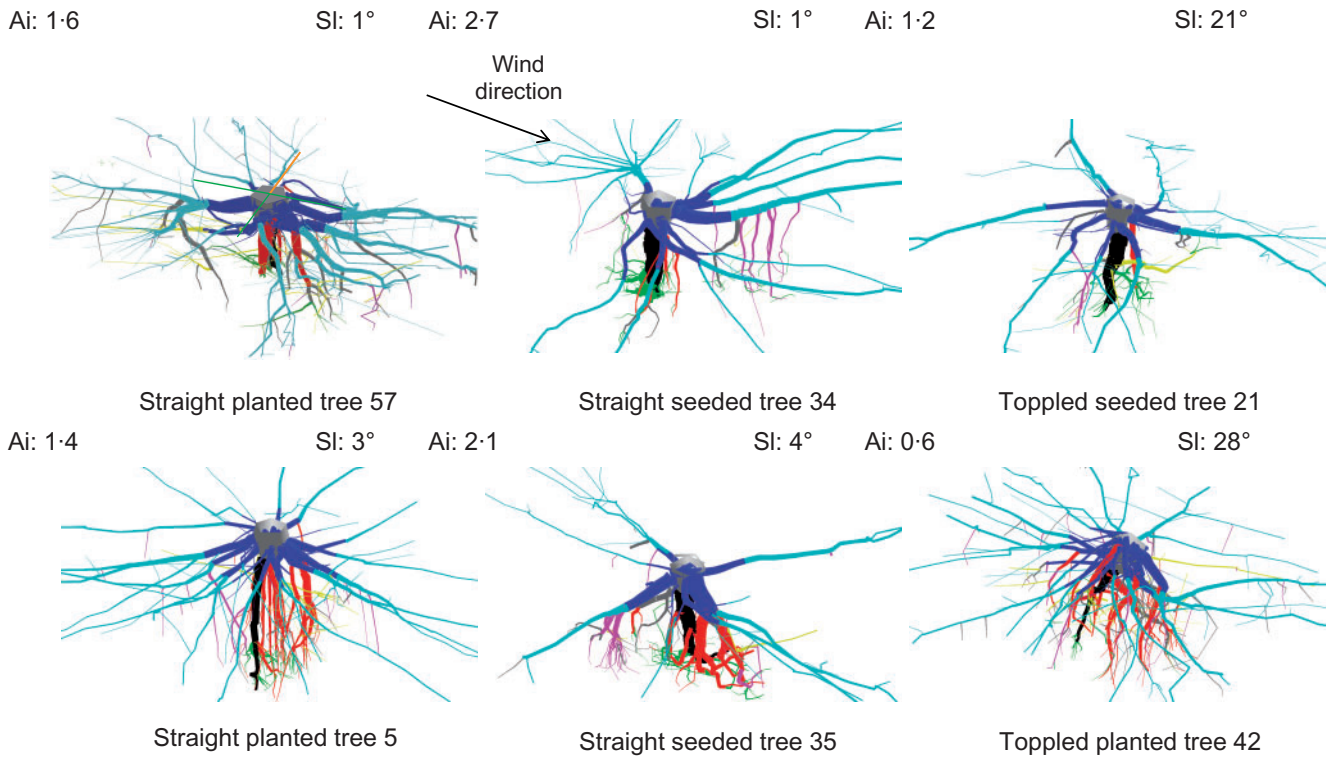


FIG. 6. 3D reconstruction of six root systems, lost roots not included. Segments were coloured according to their compartment as in Fig. 1. Same scale for all root systems. Length of the orange line within the reference frame: 50 cm. Ai, anchorage index, Sl, stem leaning.

for flexural stiffness than for maximum tensile load. Wind firmness also depends on taproot depth, which often did not reach the available soil depth in toppled trees.

A large deep-root volume is also important for anchorage because it moves the centre of rotation further to the bottom, constraining overall rotation (Ennos, 1993; Fourcaud *et al*., 2008). It lowers stresses to a point where the taproot experiences bending while its deepest part is held in tension. This may also explain why trees with a deep taproot complex did not need to reinforce their taproot structure (Fig. 5). A greater deep root volume is associated with greater flexural stiffness of the taproot, partly because the root system of woody plants more or less complies with the pipe model (Soethe *et al*., 2007): a thicker taproot at 50 cm depth has generally a larger volume of distal branches than a thinner one. The positive role of the deep root volume in anchorage was previously found in 5-year-old *P. pinaster* trees (Danjon *et al*., 1999*b*).

Menzies' taproot score mainly accounted for non-vertical growth direction and twisting of the taproot, and was associated with toppling (e.g. Mason, 1985; Toral *et al*., 2011). In this dataset, taproot vertical deviation was not related to stem leaning, probably because it did not exceed 36°, except leaning tree 56. All trees fitted in with the first two levels of the six original levels of Menzies' score, except tree 56.

In shallow soil, the tree is held by an oversized leeward shallow root or a stocky taproot with deep branches

The dispersal of straight trees on PC2 partly expressed acclimation to available soil depth. The main anchorage component of the deepest rooted tree (no. 5, Fig. 6) was a long taproot with medium flexural stiffness. A large relative root volume, stump excluded (>19 %) was needed when rooting depth was relatively shallow (<85 cm). Nicoll and Ray (1996) also found that the root mass fraction increased when rooting depth decreased. Extra biomass allocation to improve anchorage in shallow soil could be devoted to a substantial thickening of the taproot, as observed for tree 34, and to a lesser extent tree 35 (Fig. 6) which also developed a very large leeward chuck, as reported in mature trees by Danjon *et al*. (2005). Better anchorage could also be achieved by strengthening shallow root flexural stiffness leeward, through the development of one shallow root of an unusually large diameter exactly in the leeward direction. Better stem straightness has also been found for deeper-rooted trees in 5-year-old *P. pinaster* saplings (Danjon *et al*., 1999*a*).

In older *P. pinaster* (Danjon *et al*., 2005), very good acclimation to soil depth was also observed, so no relationship between soil depth and uprooting was found. Provided that available soil depth is more than 70 cm, the root system of large trees can benefit from the firm support provided by the hard pan which limits rooting depth. Smaller trees probably cannot really rest on the pan unless they develop a very strong taproot resting on the pan, with a large volume of deep roots.

Contribution of shallow roots to anchorage is mainly due to the stiffness of the end of the ZRT windward

Circular distribution is one of the main characteristics of shallow roots (Coutts *et al*., 1999). However, the windward or leeward LABS values were not relevant to explain damage,

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probably because the main component of anchorage was the taproot at that stage, and only a LABS greater than 120° is likely to weaken anchorage (Nielsen, 1990). Moreover, LABS values do not thoroughly describe the radial distribution of shallow roots. Several trees toppled right in the azimuth of one or two medium-sized roots. A 100° LABS bounded by two big roots certainly provides a better anchorage than a 50° LABS bounded by two weak roots.

Sandy soils have a high bearing capacity and low cohesion (Fourcaud *et al*., 2008). In these types of soils, during a storm, smaller leeward shallow roots cannot be pushed into the soil and therefore generally buckle early near their base, whereas windward shallow roots are more or less lifted and are later damaged by bending at a radial distance proportional to their flexural stiffness (Crook and Ennos, 1996; Fourcaud *et al*., 2008; Yang *et al*., 2014). To offset their weak taproot complex and their low deep root volume, toppled trees should have exhibited significantly stronger mechanical characteristics of their shallow roots than straight trees. But we found the opposite: flexural stiffness was even smaller windward around 2 dbh radial distance in toppled trees. This radial distance is probably the zone where roots are likely to bend or buckle. Shallow roots beyond 45 cm radial distance from the stump bark probably play a weak role in anchorage. During winching tests of *P. pinaster* trees of 15 cm dbh, Stokes (1999) did not record any strain on windward roots beyond 50 cm radial distance. In mature trees that fail by uprooting, the central part of the root system is generally rigid enough to remain undamaged; maximum tensile stress is experienced in windward roots beyond the ZRT, and maximal bending stress is experienced in leeward roots at a variable radial distance (Coutts, 1986; Nielsen, 1990; Danjon *et al*., 2005).

These root systems were digitized 3 years after the storm. When a tree leans or topples, a pronounced permanent bending moment due to self-weight appears (Alméras and Fournier, 2009), loading the shallow roots leeward and windward. The resulting acclimation may have come out as an additional diameter increment in shallow roots for 3 years, reducing the differences between straight and damaged trees for shallow root traits.

The flexural stiffness of toppled root systems is weakened by a high branching rate

The flexural stiffness of the axis of a root system decreases steeply as its diameter decreases, and decreases even more when roots branch or fork into several axes with the same total diameter (Coutts, 1983*b*). Toppled trees had finer lateral roots than straight trees. Because of higher branching, they had in particular more roots contributing to the flexural stiffness of areas most important for anchorage.

A relationship between the branching rate of shallow roots and rigidity has already stressed by Coutts (1983*b*) and Nielsen (1990). The Menzies' score modified by Gautam *et al*. (1999) also accounts for the number and forking of the taproot and main sinkers.

In straight trees only, thanks to a vertical oval shape of root cross-sections, the flexural stiffness of shallow roots increased by 5 %. Conversely, no coherent ovality pattern was noticed in the taproot complex. This difference may be explained in the following way: while most of the stress from the prevailing wind resulted in vertical flexion of shallow roots windward and leeward (Stokes, 1999), the constraints on the taproot probably had a fairly wide range of directions, as evidenced by the swaying patterns of crowns (James *et al*., 2006).

Other root system components weakly contribute to stability

Despite great variability, we found no relationship between stump relative volume and stem leaning, probably because the stump is much thicker than the surrounding roots which break first. In 10-cm-dbh *P. sitchensis*, large differences between clones in the proportion of below-ground biomass allocated either to the stump or to woody roots have been found (Nicoll *et al*., 1995).

In mature *P. pinaster*, uprooted trees diverted biomass to oblique and intermediate-depth roots probably to bring a low mechanical contribution to anchorage (Danjon *et al*., 2005; Dupuy *et al*., 2005). We did not find such a relationship in our sample, probably because the volume allocated to these root types was very low before 13 years of age.

Leeward sinkers in the ZRT contribute to anchorage by preventing leeward ZRT displacement (Crook and Ennos, 1996). In our sample, anchorage strength was certainly improved by larger leeward sinkers in tree 35 (Fig. 6), and also in a few other straight trees.

Stiffness of the central part of the root system prevents stump tilting

In small plants, efficient anchorage can be provided by a rigid taproot guyed by small shallow roots (Ennos, 1993; Weber and Mattheck, 2005; Danjon *et al*., 2013*b*). In our medium-sized trees grown in sandy soil, anchorage was mainly provided by the flexural stiffness of the central part of the root system, preventing weak displacement of the stump. Shallow roots beyond the ZRT may only marginally be loaded in tension once the stem has already leaned considerably. The distal part of long taproots and attached deep roots may be the only part of the structure experiencing great stress under tension.

Transition from failing by toppling to failing by uprooting, at the pole stage

In our straight pole-stage trees, the most efficient root system components for anchorage, namely the taproot and deep roots, averaged less than 4 % of the stem volume, and 6 % when other roots from the taproot complex were included. Conversely, the shallow root volume averaged nearly 15 % of the stem volume and displayed high acclimation to the prevailing wind direction, even if these roots apparently contributed only moderately to anchorage. Despite a lower actual role in anchorage, good acclimation to prevailing wind in the circular distribution of shallow roots has already been noted in these 13-year-old trees. This can be explained in two ways: (1) a greater part of the root biomass was devoted to the soil horizons with higher productivity (Korndörfer *et al*., 2008), and (2) the trees had started the

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transition between the juvenile stage when they fail by toppling to the adult stage when they fail by uprooting. Thus, the trees were ready to build the major element of their root cage, a rigid ZRT, as well as the shallow root beyond ZRT structure guying this cage. During this transition phase, from a mechanical point of view, the trees had neither the most efficient taproot system nor the most efficient secondary sinker root system. The shallower the soil, the more rapidly this transition has to start (Ennos, 1993).

Although all toppled trees had a taproot, they did not display sufficient plasticity to allocate a percentage of their biomass to strengthening the taproot complex or at least strongly thickening their ZRT roots windward or leeward. As noticed by Nicoll *et al*. (1995), biomass allocation to root system components and root architecture are partly under genetic control. Accordingly, 30 % of the variability of stem basal sweep in *P. pinaster* is controlled by genetics, so that differences in biomass allocation to roots could be due to different growth strategies (Danjon, 1994). *Pinus pinaster* from different biotopes exhibit considerable differences in root architecture, attributed to natural selection, whether for wind-firmness or for drought avoidance (Danjon *et al*., 2009).

The predominance of flexural stiffness implies that anchorage strength is not only governed by the spatial distribution of biomass allocation to roots, but also by other architectural features such as root number, heterogeneity of diameter distribution and cross-section shape (Coutts *et al*., 1999). The mechanically active part of the juvenile to mature *P. pinaster* root framework is mainly composed of the roots grown during the first years in the field. This implies that the root system has to show large plasticity in diameter growth to build a mechanically efficient structure for all stages of the tree's life and to compensate for missing components of the root system, if required.

We could unravel anchorage mechanisms thanks to new analysis techniques

First, even if the size of our sample was rather small, we levelled out differences in development stage or in aerial part size by sampling trees of similar sizes, and we also got rid of the potential bias due to lost roots. We chose to perform a very detailed measurement of root architecture, and computed variables that matched best with data from the literature about the mechanical functioning of root systems. The analysis methods built up by Danjon *et al*. (2005, 2013*b*) yielded only few differences among treatments. For example, examining the circular distribution of shallow roots by defining windward and leeward 90°-wide quadrants (e.g. Danjon *et al*., 2005) resulted in a very large threshold effect, which was overridden by our weighting procedure. In the same way, computing only root volumes or CSAs provided partial information on flexural stiffness.

Practical recommendations

To further prevent anchorage failure at the sapling stage, nursery and re-afforestation techniques should first promote the growth of a deep, unique, almost vertical and large taproot. Secondly, windward and leeward shallow roots should progressively form a structure with great flexural stiffness, but tending machines should not cut living tree roots in the vicinity of the stump. The lack of interaction between leaning and regeneration indicates that there was no fundamental difference in anchorage functioning between planted and seeded trees of the same size. Nevertheless, at the stand scale, more trees toppled in the planted stands than in the seeded stand.

CONCLUSION

Root architecture appears to be a major factor determining anchorage failure. Thanks to detailed measurements and a thorough analysis of root architecture, including classification by root types and computation of mechanical characteristics in a fairly large sample of trees of the same size, we can now better understand which components of the root system prevent anchorage failure in young taprooted trees.

From an experimental point of view, stem straightness, root architecture measurements and tree pulling experiments should be done periodically up to the pole stage to determine how various factors influence the architectural development of the root system and the related anchorage strength. Trials should target genetic units, soil types or installation methods.

To test our hypotheses, to determine the dynamics of interactions between root components, and to better assess the role of root geometry, a biomechanical model like RootAnchor (Yang *et al*., 2014) could be used. The model could be based on root system measurements, root system growth models (e.g. Pagès *et al*., 2004) or schematic root patterns corresponding to typical architectural features.

SUPPLEMENTARY DATA

Supplementary data are available online at www.aob.oxfordjournals.org and consist of the following. Table S1: values of the variables for each tree.

ACKNOWLEDGEMENTS

We thank Raphaël Segura and Nesrine Mohamed Omar (INRA, UMR1202 BioGeCo), Jean-Mathieu de Boisseson, Francis Melun, Ingrid Bastard and Pierre Gauthier (FCBA, Pôle Biotechnologie Sylviculture Avancée), Pierre Trichet, Didier Garrigou, Jean-Marc Bonnefond and Dominique Guyon (INRA, UMR1391 ISPA), and Nicolas Dubreuil (INRA, UE0570, Unité Expérimentale Forêt Pierroton) for technical support and for their helpful assistance with fieldwork. We thank Sébastien Cavaignac, Alain Bailly and Ambre Le Ferrec (FCBA, Pôle Biotechnologie Sylviculture Avancée) for managing the research project, and Pauline Défossez and Ming Yang (INRA, UMR1391 ISPA) for managing the companion winching study and for useful comments. We also thank the Caisse Des Dépôts et Consignations and the Société Forestière for allowing us to work on the stands and for assistance (Ceydric Sedilot-Gasmi, Guillaume Silande and Bernard Lagardère). The PhD grant of Antoine Danquechin Dorval was financed by the Caisse Des Dépôts et Consignations and the Conseil Régional d'Aquitaine

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(Diademe project - 2011 1203004). The main funding was provided by the Caisse Des Dépôts et Consignations. The companion study was supported by the Conseil Régional d'Aquitaine and the French Ministère de l'Agriculture, de l'Agroalimentaire et de la Forêt (Fortius project – 13001087).

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